

Observations:-

Table No 01 :- Experimental data of the time taken to blueness at different temperatures.

| Step | Temperature / °C | Time (s) |
|------|------------------|----------|
| (a)  | 35               | $t_i$    |
|      | 30               |          |
|      | 25               |          |
|      | 20               |          |
|      | 15               |          |
| (b)  | 25               |          |
| (c)  | 25               |          |

Calculation :-

(e). Initial  $n_{S_2O_8^{2-}} = 0.002 \text{ mol dm}^{-3} \times \frac{10}{1000} \text{ dm}^3$   
 $= 0.2 \times 10^{-4} \text{ mol}$

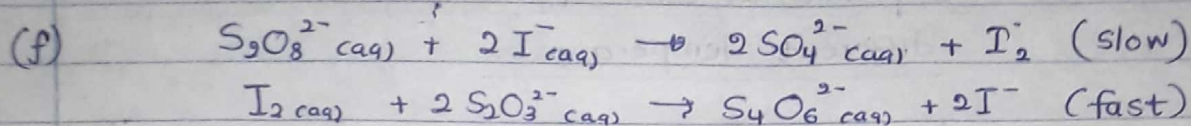
Initial  $[S_2O_8^{2-}] = \frac{0.2 \times 10^{-4} \text{ mol dm}^{-3}}{20 \times 10^{-3}}$   
 $= 0.001 \text{ mol dm}^{-3} //$

Initial  $n_{I^-} = 0.1 \text{ mol dm}^{-3} \times \frac{5}{1000} \text{ dm}^3$

$= 5 \times 10^{-4} \text{ mol}$   
 Initial  $[I^-] = \frac{5 \times 10^{-4} \text{ mol dm}^{-3}}{20 \times 10^{-3}}$   
 $= 0.025 \text{ mol dm}^{-3}$

for (a)

Likewise you should calculate the initial concentration of  $S_2O_8^{2-}$  and  $I^-$  in mixture (b) and (c)



$$n_{S_2O_8^{2-}} : n_{S_4O_6^{2-}}$$

$$1 : 2$$

$$n_{S_2O_3^{2-}} = 5 \times 10^{-3} \text{ dm}^3 \times 0.0004 \text{ mol dm}^{-3}$$

$$= 2 \times 10^{-6} \text{ mol}$$

$$\text{Reacted } S_2O_8^{2-} = \frac{2 \times 10^{-6}}{2} \text{ mol} = 1 \times 10^{-6} \text{ mol}$$

$$\text{Initial } n_{S_2O_8^{2-}} = 0.002 \times 10 \times 10^{-3} = 0.2 \times 10^{-4} \text{ mol}$$

$$\text{after (t) time } n_{S_2O_8^{2-}} = (0.02 - 0.00) \times 10^{-3}$$

$$= 0.19 \times 10^{-4} \text{ mol}$$

$$[S_2O_8^{2-}] = \frac{0.19 \times 10^{-4} \text{ mol}}{20 \times 10^{-4} \text{ dm}^3} = \frac{9.5 \times 10^{-4}}{20} \text{ mol dm}^{-3}$$

$$= 0.00475 \text{ mol dm}^{-3}$$

$$\text{Initial rate} = \frac{-(C_t - C_0)}{t}$$

$$= \frac{-(0.00475 - 0.001) \text{ mol dm}^{-3}}{t_1}$$

(This is for (a) at 25°C)

Likewise you should calculate the initial rate of reaction for (b) and (c) at 25°C and (a) for 15°C, 20°C, 30°C and 35°C



(9) Table No 02 :- Initial rate of the reaction of a mixture at 25°C.

| Mixture | Initial rate $\text{mol dm}^{-3} \text{s}^{-1}$ |
|---------|-------------------------------------------------|
| (a)     | rate <sub>1</sub>                               |
| (b)     | rate <sub>2</sub>                               |
| (c)     | rate <sub>3</sub>                               |

$$\text{rate} = k [\text{S}_2\text{O}_8^{2-}]^n [\text{I}^-]^m$$

for mixture (a)

$$\text{rate}_1 = k [C_1]^n [C_2]^m \quad \text{--- (1)}$$

for mixture (b)

$$\text{rate}_2 = k [C_3]^n [C_4]^m \quad \text{--- (2)}$$

for mixture (c)

$$\text{rate}_3 = k [C_5]^n [C_6]^m \quad \text{--- (3)}$$

(By using 2 equations you can calculate the n and m)

$$n = ?$$

$$m = ?$$

(1).

Determine the rate constant at different temperatures:-

at 15°C

$$k_1 = \frac{\text{Rate}}{[\text{S}_2\text{O}_8^{2-}]^n [\text{I}^-]^m}$$

you know n and m,  
So you can calculate the  
 $k_1$  value

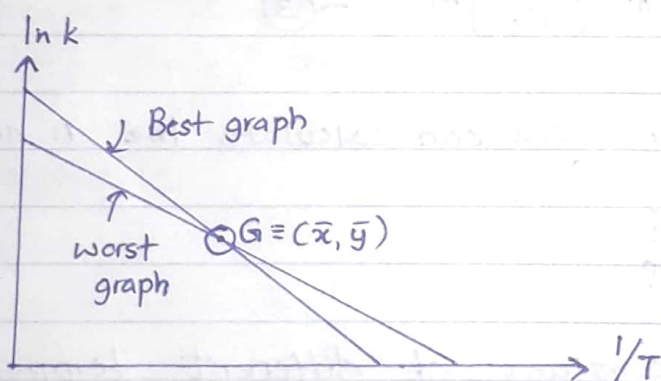
Likewise calculate the k values for 20°C, 25°C, 30°C, 35°C

Table No 03 :- Data sheet for  $\ln k$  vs  $1/T$  graph.

| Temperature<br>( $\pm 0.5$ ) / K | Rate constant<br>$\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$ | $\ln k$<br>y axis | $1/T$<br>( $\text{K}^{-1}$ ) x axis |
|----------------------------------|--------------------------------------------------------------|-------------------|-------------------------------------|
| 273 + 15                         |                                                              |                   |                                     |
| 273 + 20                         |                                                              |                   |                                     |
| 273 + 25                         |                                                              |                   |                                     |
| 273 + 30                         |                                                              |                   |                                     |
| 273 + 35                         |                                                              |                   |                                     |

G point calculation

$$G \equiv (\bar{x}, \bar{y})$$



$$\ln k = \frac{-E_a}{R} \times \frac{1}{T} + \ln A$$

$$m_{\text{best}} =$$

$$m_{\text{worst}} =$$

$$m_{\text{error}} =$$

$$(E_a)_{\text{best}} =$$

$$(E_a)_{\text{worst}} =$$

$$(E_a)_{\text{error}} =$$

$$E_a = (E_a)_{\text{best}} \pm (E_a)_{\text{error}} //$$

Conclusion :-

The order of the reaction with respect to the  $S_2O_8^{2-}(aq)$  concentration is  $(n) = \dots\dots\dots$

The order of the reaction with respect to the  $I^-(aq)$  concentration is  $(m) = \dots\dots\dots$

The overall order of the reaction is  $\dots\dots\dots$

The activation energy ( $E_a$ ) of the reaction is  $\dots\dots\dots$

Discussion :-

- All the chemical reactions
- chemistry behind the practical
- Rate constant
- order of the reaction.
- Reason for adding starch
- Why  $S_2O_8^{2-}$  adding finally.

References :-